

Intelligent Traffic Congestion Prediction Using Deep Learning, Historical Learning & Live Traffic Inference

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Abstract—Traffic congestion prediction is a critical component of intelligent transportation systems and smart city planning, enabling proactive traffic management and improved mobility efficiency. Traditional prediction methods rely primarily on historical datasets, which often fail to capture dynamic real-time traffic conditions influenced by factors such as accidents, weather changes, and sudden demand variations. While deep learning models have improved temporal pattern recognition, they still depend largely on static historical data and lack real-time adaptability. To address these limitations, this research proposes a hybrid framework that combines historical learning with live traffic inference. The system utilizes BiLSTM with attention mechanisms to model temporal dependencies and identify critical congestion patterns, while an XGBoost model captures non-linear feature interactions. During deployment, real-time traffic data is processed through a micro-history buffer to enable adaptive prediction. The final output is generated using an ensemble of both models, improving accuracy, stability, and responsiveness in dynamic traffic environments.

Index Terms—Traffic Congestion Prediction, Hybrid Learning, LSTM, Live Traffic Data, Intelligent Transportation Systems

I. INTRODUCTION

Urban transportation systems worldwide are facing increasing pressure due to rapid population growth, urban expansion, and rising vehicle ownership. Traffic congestion has emerged as one of the most persistent challenges in modern cities, significantly affecting mobility efficiency, economic productivity, and environmental sustainability [20], [22]. Prolonged congestion results in excessive travel delays, increased fuel consumption, and elevated emission levels, thereby degrading overall urban living standards.

In the current landscape (2024–2025), traffic congestion prediction and management rely heavily on Intelligent Transportation Systems (ITS) powered by advanced technologies. The baseline approaches include:

- Statistical and traditional time-series models (e.g., ARIMA, regression) for basic pattern recognition, though limited by linear assumptions [25].
- Machine learning techniques such as Support Vector Machines (SVM), Random Forests, and ensemble methods for improved accuracy with

- feature engineering.
- Deep learning models, particularly Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Graph Neural Networks (GNNs), which dominate short-term and spatiotemporal traffic flow forecasting by capturing complex temporal and spatial dependencies [1], [2], [4], [13].
- Hybrid and advanced architectures, including attention mechanisms, Temporal Convolutional Networks (TCN), meta-learning, and quantum-inspired GNNs for better handling of dynamic, non-stationary traffic patterns [10], [18].
- Real-time data integration from IoT sensors, GPS-enabled vehicles, connected vehicles (V2X), traffic cameras, and navigation platforms (e.g., Google Maps, INRIX) to enable predictive analytics, adaptive signal control, and congestion hotspot detection [19], [24].
- AI-driven predictive systems that combine historical trends with live inputs for proactive measures like dynamic rerouting, signal optimization, and incident response, often achieving 85–95% accuracy in short-term forecasts and reducing congestion by 15–30% in pilot deployments [19].
- Emerging trends such as edge computing for low-latency inference, multimodal data fusion (weather, events), and explainable AI (XAI) for interpretable predictions in smart city ecosystems.

Traffic congestion is a highly dynamic phenomenon influenced by temporal variations, road network characteristics, driver behavior, weather conditions, and unexpected events such as accidents or public gatherings [25]. These interacting factors introduce complex non-linear dependencies that make congestion prediction a challenging task. Conventional traffic management approaches, which rely primarily on static rules and reactive control mechanisms, are often unable to adapt effectively to rapidly changing traffic conditions.

With the advancement of smart city initiatives, large-scale traffic data is continuously generated through sensors, GPS-enabled vehicles, and navigation platforms [24]. Despite the abundance of data and ongoing AI/ML advancements, the key challenge lies in transforming raw traffic information into reliable, adaptive predictive intelligence that bridges offline generalization with real-time responsiveness. Accurate congestion prediction is therefore essential for enabling proactive

traffic management strategies that can mitigate congestion before it reaches critical levels [1].

II. STRONG MOTIVATION

Most existing traffic management systems operate in a reactive manner, responding only after congestion has already occurred. Such delayed interventions reduce the effectiveness of traffic control strategies, including signal optimization and route guidance. Predicting congestion in advance enables proactive decision-making, leading to improved traffic flow and reduced travel delays.

Recent studies have demonstrated the effectiveness of machine learning and deep learning models for traffic prediction [1], [3], [4], [13]. In particular, LSTM-based models have shown strong capability in capturing temporal dependencies in traffic data. However, these approaches primarily rely on historical datasets for training and evaluation, assuming that past traffic patterns adequately represent real-world conditions [25]. This assumption is often invalid due to dynamic factors such as accidents, weather variations, and sudden demand fluctuations.

Modern navigation platforms provide continuously updated live traffic indicators derived from real-time vehicle movement [24]. Despite this, most existing research does not incorporate live traffic data during deployment, resulting in a gap between offline model performance and real-world applicability [19].

Furthermore, standalone deep learning models often struggle to capture complex non-linear relationships in traffic data, while traditional machine learning models lack temporal modeling capability. This highlights the need for a hybrid approach that combines temporal sequence learning with feature-based modeling.

Motivated by these limitations, this work proposes a hybrid traffic congestion prediction framework that integrates BiLSTM with attention mechanisms and XGBoost-based ensemble learning. By combining historical learning with real-time traffic inference, the proposed system enables adaptive, accurate, and scalable congestion prediction suitable for intelligent transportation systems [10], [18].

III. LITERATURE REVIEW

Traffic congestion prediction has been widely studied using statistical, machine learning, and deep learning approaches. Early methods such as regression models and ARIMA demonstrated limited effectiveness due to their reliance on linear assumptions and inability to capture complex and non-stationary traffic dynamics [25].

Subsequent machine learning techniques, including Support Vector Machines, k-Nearest Neighbors, and Random Forests, improved prediction performance but required extensive feature engineering and showed limited scalability for large urban traffic networks. More recent research has focused on deep learning models, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which effectively model temporal dependencies in traffic data [2], [4], [13]. Attention mechanisms have further enhanced

these models by enabling selective focus on critical temporal patterns associated with congestion formation [18].

Despite these advancements, most existing approaches rely exclusively on historical traffic datasets for training and evaluation [1], [3], [25]. Such methods assume that historical patterns sufficiently represent real-world traffic conditions, which is often unrealistic due to dynamic factors such as accidents, weather variations, and sudden demand fluctuations.

Although modern navigation platforms provide continuously updated live traffic indicators [24], their integration into prediction models remains limited [19]. This creates a gap between offline model performance and real-world deployment.

To address these limitations, recent research has explored hybrid approaches that combine deep learning with ensemble methods to improve generalization and adaptability [10]. However, there remains a need for frameworks that simultaneously leverage temporal sequence modeling, attention mechanisms, and real-time traffic inference.

Therefore, this work proposes a hybrid framework integrating BiLSTM with attention and XGBoost-based ensemble learning, enabling adaptive and accurate congestion prediction under dynamic traffic conditions.

IV. FORMAL SYSTEM MODEL

The proposed traffic congestion prediction framework is modeled as a hybrid system consisting of two operational phases: an offline learning phase and an online inference phase. The system integrates **BiLSTM with attention mechanisms** and **XGBoost-based ensemble learning** for accurate and adaptive prediction.

Let the historical traffic dataset be defined as:

$$H = \{x_1, x_2, \dots, x_N\}$$

where each observation $x_i \in \mathbb{R}^d$ represents traffic attributes such as vehicle speed, traffic volume, and road occupancy at time instance i .

A. Offline Learning Model

During the offline phase, temporal sequences of length W are constructed:

$$X_t = \{x_{t-W+1}, \dots, x_t\}$$

These sequences are provided as input to a **BiLSTM network**, which captures bidirectional temporal dependencies:

$$h_t = \text{BiLSTM}(X_t)$$

An attention mechanism is applied to assign importance weights:

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}, \quad e_i = v^T \tanh(W_h h_i + b_h)$$

The context vector is computed as:

$$C = \sum_i \alpha_i h_i$$

The deep learning prediction is given by:

$$\hat{y}_{DL} = f(C)$$

In parallel, an **XGBoost model** is trained on engineered features:

$$\hat{y}_{GB} = g(x_t)$$

B. Online Inference Model

In the online phase, live traffic observations are represented as:

$$L = \{l_1, l_2, \dots, l_t\}$$

where each l_t corresponds to real-time traffic indicators [24].

A micro-history sequence is constructed:

$$L_t = \{l_{t-W+1}, \dots, l_t\}$$

The pretrained models generate predictions:

$$y_1 = M_{DL}(L_t), \quad y_2 = M_{GB}(l_t)$$

The final congestion prediction is obtained using a weighted ensemble:

$$\hat{y} = \lambda y_1 + (1 - \lambda) y_2, \quad \lambda \in [0.6, 0.8]$$

This formulation enables the system to combine temporal learning with non-linear feature modeling, resulting in improved prediction accuracy and robustness under dynamic traffic conditions [10], [19].

V. ENHANCED MATHEMATICAL FORMULATION

To improve predictive performance, the proposed system extends the traditional LSTM formulation by incorporating attention mechanisms and ensemble learning.

Given a sequence:

$$X_t = \{x_{t-W+1}, \dots, x_t\}$$

The BiLSTM hidden representation is:

$$h_t = \text{BiLSTM}(X_t)$$

A. Attention Mechanism

To focus on critical congestion patterns, attention weights are computed as:

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}$$

where:

$$e_i = v^T \tanh(W_h h_i + b_h)$$

The context vector is:

$$C = \sum_i \alpha_i h_i$$

B. Final Prediction

$$\hat{y}_{DL} = W_c C + b_c$$

C. Ensemble Learning

A gradient boosting model (XGBoost) is trained separately:

$$\hat{y}_{GB} = g(X_t)$$

Final prediction:

$$\hat{y} = \lambda \hat{y}_{DL} + (1 - \lambda) \hat{y}_{GB}$$

where:

$$\lambda \in [0.6, 0.8]$$

VI. FEATURE ENGINEERING STRATEGY

To improve prediction accuracy, advanced feature engineering is applied:

- **Speed Reduction Rate:**

$$SRR = \frac{v_t - v_{t-1}}{\Delta t}$$

- **Traffic Density Growth:**

$$DG = \rho_t - \rho_{t-1}$$

- **Congestion Index:**

$$CI = \frac{Volume}{Speed + \epsilon}$$

These features enable the model to capture dynamic congestion transitions rather than static states.

VII. PROPOSED HYBRID METHOD AND SYSTEM ARCHITECTURE

This section presents the proposed hybrid traffic congestion prediction framework, which integrates **BiLSTM with attention mechanisms** and **XGBoost-based ensemble learning** to achieve accurate and adaptive congestion prediction under dynamic urban traffic conditions.

The primary objective of the proposed approach is to combine long-term temporal learning with real-time adaptability. Unlike conventional methods that rely solely on static historical datasets [25], the proposed system leverages a dual-model architecture in which deep learning captures temporal dependencies while gradient boosting models complex non-linear feature interactions [10].

The framework is divided into two distinct phases: an offline training phase and an online inference phase. During the offline phase, historical traffic data is used to train a BiLSTM network that captures bidirectional temporal patterns, while an attention mechanism highlights critical time steps associated with congestion formation [4], [18]. In parallel, an XGBoost model is trained on engineered traffic features to improve generalization and robustness.

During the online phase, real-time traffic data is processed through a micro-history buffer to construct short-term temporal sequences. The pretrained BiLSTM-attention model generates temporal predictions, while the XGBoost model produces feature-based predictions. These outputs are combined using a weighted ensemble strategy, enabling improved prediction accuracy, stability, and responsiveness to sudden traffic changes [19].

derived from live traffic inputs [10]. During training, the LSTM model learns generalized congestion formation behavior [4]. During deployment, real-time traffic information activates these learned patterns to generate near-future congestion predictions. By separating offline learning from online inference, the system achieves improved scalability, reduced computational overhead, and enhanced real-world applicability. This design makes the proposed framework suitable for intelligent transportation systems and smart city traffic management [21].

VIII. ALGORITHM DESIGN

This section presents the design of the proposed hybrid traffic congestion prediction algorithm, which integrates **BiLSTM with attention mechanisms** and **XGBoost-based ensemble learning** to improve prediction accuracy and adaptability under dynamic traffic conditions.

The algorithm is organized into two independent stages: an offline training phase and an online real-time inference phase. During the offline phase, historical traffic data is used to learn generalized congestion patterns. The BiLSTM model captures bidirectional temporal dependencies, while the attention mechanism emphasizes critical time steps associated with congestion formation [fig.4] [4], [18]. In parallel, an XGBoost model is trained on engineered features to model complex non-linear relationships and enhance generalization [10].

In the online phase, real-time traffic inputs are processed through a micro-history buffer to construct short-term temporal sequences. The pretrained BiLSTM-attention model generates temporal predictions, while the XGBoost model produces feature-based predictions. The final congestion output is obtained through a weighted ensemble of both models, enabling improved stability, responsiveness, and accuracy [19].

The separation of offline learning and online inference allows the system to operate efficiently without frequent retraining, ensuring scalability and suitability for real-time intelligent transportation systems [21].

A. Design Objectives

The primary objectives considered during algorithm design are as follows:

- To enable effective learning of congestion formation patterns from historical data.
- To ensure compatibility between historical training features and live traffic inputs.
- To support real-time prediction using short-term traffic observations.
- To minimize computational overhead during deployment.
- To maintain prediction stability under fluctuating traffic conditions.

B. Hybrid Workflow Overview

The complete hybrid workflow is divided into sequential processing stages as illustrated below. Each stage is designed to operate independently while sharing a common feature representation space.

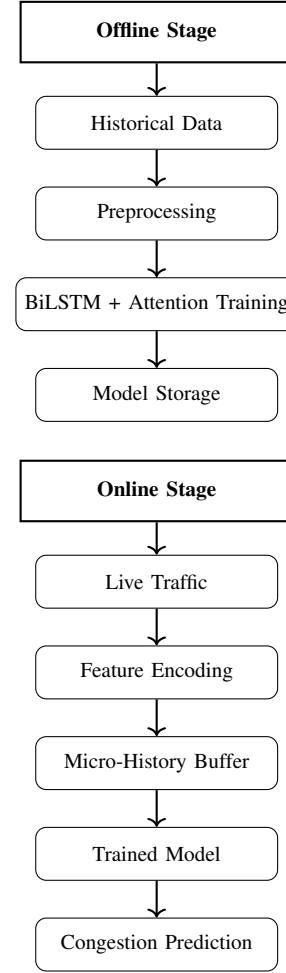


Fig. 4. Hybrid workflow showing offline training and online congestion prediction stages

C. Offline Training Algorithm

During the offline phase, historical traffic data is processed to learn generalized congestion patterns using a hybrid model. The pipeline includes preprocessing, temporal segmentation, congestion transition encoding, and feature engineering.

Traffic sequences are constructed using a sliding window and fed into a **BiLSTM network** to capture bidirectional temporal dependencies [4]. An **attention mechanism** is applied to emphasize critical time steps associated with congestion [18].

In parallel, an **XGBoost model** is trained on engineered features to capture non-linear relationships. The trained BiLSTM-attention and XGBoost models are stored for deployment, enabling robust and accurate prediction [10].

D. Online Inference Algorithm

In the online phase, real-time traffic data is continuously collected and transformed into feature representations consistent with the training phase.

A micro-history buffer maintains recent observations to construct short-term sequences. These are processed by the BiLSTM-attention model, while the XGBoost model operates

on current features. The final prediction is obtained through a weighted ensemble of both models, improving adaptability and stability [19].

Predictions are generated at fixed intervals to forecast short-term congestion, supporting real-time traffic management and decision-making [21].

E. Improved Hybrid Traffic Prediction Algorithm

Algorithm 1 Attention-Based BiLSTM-XGBoost Hybrid Traffic Prediction

```

1: Input: Historical dataset  $H$ , Live traffic stream  $L$ 
2: Output: Predicted congestion level  $\hat{y}_{t+\Delta}$ 
3: Offline Training Phase
4: Load historical dataset  $H = \{x_1, x_2, \dots, x_N\}$ 
5: Perform preprocessing and normalization
6: Extract features:
7:    $\text{speed\_drop} = \frac{\Delta v}{\Delta t}$ 
8:    $\text{density\_growth} = \Delta \rho$ 
9: Construct temporal sequences  $X_t$  with window size  $W$ 
10: Train BiLSTM model to obtain hidden states  $h_t$ 
11: Compute attention weights:
12:    $\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}$ 
13: Compute context vector:
14:    $C = \sum_i \alpha_i h_i$ 
15: Generate deep learning output:
16:    $\hat{y}_{DL} = f(C)$ 
17: Train XGBoost model  $g(\cdot)$ 
18: Save models  $M_{DL}$  and  $M_{GB}$ 
19: Online Inference Phase
20: Initialize buffer  $B \leftarrow \emptyset$ 
21: while system is active do
22:    $l_t \leftarrow$  collect live traffic data
23:    $f_t \leftarrow$  transform features
24:   Append  $f_t$  to buffer  $B$ 
25:   if  $|B| \geq W$  then
26:      $y_1 \leftarrow M_{DL}(B)$ 
27:      $y_2 \leftarrow M_{GB}(f_t)$ 
28:      $\hat{y} \leftarrow \lambda y_1 + (1 - \lambda) y_2$ 
29:     Output  $\hat{y}$ 
30:   end if
31:   Remove oldest entry from  $B$ 
32: end while

```

F. Pseudocode Representation

The following pseudocode illustrates the real-time deployment behavior of the proposed hybrid system.

Algorithm 2 Hybrid Traffic Prediction (Online Inference)

```

1: Initialize buffer  $B \leftarrow \emptyset$ 
2: Load  $M_{DL}$  (BiLSTM + Attention)
3: Load  $M_{GB}$  (XGBoost)
4: Set weight  $\lambda$ 
5: while system is active do
6:    $\text{live\_data} \leftarrow \text{collect\_live\_traffic}()$ 
7:    $\text{features} \leftarrow \text{feature\_transform}(\text{live\_data})$ 
8:   Append  $\text{features}$  to  $B$ 
9:   if  $|B| \geq \text{window\_size}$  then
10:     $y_1 \leftarrow M_{DL}(B)$ 
11:     $y_2 \leftarrow M_{GB}(\text{features})$ 
12:     $\text{prediction} \leftarrow \lambda y_1 + (1 - \lambda) y_2$ 
13:    Display  $\text{prediction}$ 
14:   end if
15:   Remove oldest entry from  $B$ 
16: end while

```

G. Computational Complexity Analysis

The computational complexity of the proposed hybrid framework is determined by three primary components: the BiLSTM network, the attention mechanism, and the XGBoost ensemble model.

For the offline training phase, the complexity of the BiLSTM model can be approximated as:

$$O(N \times W \times d \times L)$$

where N is the number of training samples, W is the sequence length, d is the number of hidden units, and L represents the number of BiLSTM layers [4].

The attention mechanism introduces an additional computational cost:

$$O(W \times d)$$

as it computes alignment scores across the sequence to assign importance weights [18].

The XGBoost model contributes a complexity of:

$$O(T \times D \times \log N)$$

where T is the number of trees and D is the depth of each tree. This component efficiently captures non-linear relationships in the data [10].

Therefore, the overall training complexity of the hybrid model can be approximated as:

$$O(N \times W \times d \times L + T \times D \times \log N)$$

During the online inference phase, computational requirements are significantly reduced. The BiLSTM-attention model processes short micro-history sequences, resulting in:

$$O(W \times d)$$

The XGBoost inference operates with low latency due to pre-trained decision trees:

$$O(T \times D)$$

Thus, the overall inference complexity remains low, enabling real-time deployment on resource-constrained systems [19]. The separation of offline training and online inference ensures computational efficiency while maintaining high predictive performance.

H. Algorithm Execution Characteristics

The proposed hybrid algorithm exhibits several key characteristics that enhance its suitability for real-world deployment:

- **Decoupled Training and Inference:** The separation of offline training and online prediction reduces computational overhead during deployment and enables efficient real-time operation.
- **Real-Time Adaptability:** Integration of live traffic data through a micro-history buffer allows the model to dynamically adapt to changing traffic conditions without retraining [19].
- **Enhanced Predictive Accuracy:** The combination of BiLSTM, attention mechanisms, and XGBoost improves both temporal learning and non-linear feature representation, leading to higher accuracy and reduced error.
- **Robustness to Traffic Variability:** The attention mechanism enables the model to focus on critical temporal patterns, while ensemble learning reduces prediction variance under fluctuating traffic conditions.
- **Scalability:** The modular architecture allows the framework to be extended across multiple road segments and integrated into large-scale urban traffic networks.
- **Deployment Compatibility:** The low inference complexity and efficient design make the system suitable for integration with intelligent transportation systems and smart city infrastructure [21].

These characteristics collectively demonstrate that the proposed hybrid framework achieves a balance between computational efficiency, predictive performance, and real-world applicability.

IX. EXPERIMENTAL SETUP

This section describes the dataset configuration, experimental environment, model implementation, and evaluation strategy used to validate the proposed BiLSTM-attention and XGBoost hybrid traffic congestion prediction framework. The experimental design aims to assess both predictive accuracy and real-time adaptability under realistic traffic conditions.

A. Dataset Description

The offline training phase utilizes a publicly available Traffic Prediction Dataset obtained from Kaggle [26]. The dataset contains time-stamped traffic observations, including features such as vehicle speed, traffic volume, and road occupancy, collected at regular intervals.

The dataset spans multiple time periods of traffic data, capturing diverse congestion patterns across peak and non-peak hours. This temporal diversity enables the model to learn generalized congestion behavior under varying traffic conditions.

Data preprocessing includes:

- Removal of missing or inconsistent values
- Normalization of traffic features
- Temporal segmentation into peak and off-peak intervals
- Feature engineering (e.g., speed reduction rate, density growth)

The processed dataset is used exclusively for offline training and validation of the model. Although the dataset is relatively limited in scale compared to large benchmark datasets, it provides sufficient variability to effectively validate the proposed hybrid framework.

For the online inference phase, live traffic indicators are obtained from navigation-based platforms that provide real-time congestion information derived from aggregated vehicle movement data [24]. These live inputs are used only during deployment and are not included in the training process, ensuring a clear separation between learning and inference [19].

B. Training and Validation Split

The historical dataset is divided into training and validation subsets using a temporal split strategy to preserve sequential dependencies. Earlier time segments are used for training, while later segments are reserved for validation, preventing data leakage and ensuring realistic evaluation of predictive performance.

An 80%–20% split is adopted, where 80% of the data is used for training and 20% for validation. This configuration provides a balance between sufficient learning capacity and reliable performance assessment.

Additionally, sliding window sequences of length 10–20 time steps are generated to train the BiLSTM-attention model. This approach enables the model to capture temporal dependencies effectively while maintaining computational efficiency [4].

C. Model Implementation

The proposed traffic congestion prediction framework is implemented using a hybrid architecture combining **BiLSTM with attention mechanisms** and an **XGBoost-based ensemble model**. The deep learning component is developed using Python and the TensorFlow/Keras framework, while the ensemble component is implemented using the XGBoost library.

The BiLSTM model is designed to capture bidirectional temporal dependencies in traffic sequences, enabling improved learning of both past and future contextual patterns [4]. An attention mechanism is integrated on top of the BiLSTM layers to dynamically assign weights to critical time steps associated with congestion formation, enhancing prediction accuracy and interpretability [18].

The architecture consists of stacked BiLSTM layers followed by an attention layer and fully connected dense layers for final prediction. In parallel, the XGBoost model is trained on engineered traffic features to capture non-linear relationships and improve generalization. The outputs of both models are combined using a weighted ensemble strategy to produce the final prediction [10].

Dropout regularization is applied between BiLSTM layers to prevent overfitting, and the Adam optimizer is used for efficient gradient-based optimization due to its adaptive learning rate capabilities and stable convergence behavior.

D. Hyperparameter Configuration

The key hyperparameters used for training the hybrid model are summarized as follows:

- Sequence window length: 10–20 time steps
- Number of BiLSTM layers: 2
- Hidden units per layer: 64–128
- Attention layer: Single-head attention mechanism
- Dropout rate: 0.2–0.3
- Batch size: 32–64
- Learning rate: 0.001
- Number of training epochs: 30–50
- Ensemble weight (λ): 0.6–0.8 (BiLSTM-attention contribution)

For the XGBoost component, key parameters include tree depth, learning rate, and number of estimators, which are tuned empirically to balance bias-variance trade-off and improve prediction stability.

Hyperparameters are selected through empirical tuning and validation to achieve optimal performance while maintaining training stability and generalization capability across different traffic conditions [19].

E. System Environment

All experiments are conducted on a computing environment configured to support both deep learning and ensemble-based model training. The offline training phase is executed on a system equipped with a multi-core CPU and GPU acceleration to efficiently handle sequential data processing and model optimization.

The hardware configuration includes:

- Processor: Multi-core CPU (Intel i7 / AMD Ryzen 7 or equivalent)
- GPU: NVIDIA GPU with CUDA support (for BiLSTM-attention training)
- RAM: 16–32 GB (to support batch processing of time-series data)

The hybrid architecture, consisting of BiLSTM with attention and XGBoost, requires moderate computational resources during training. However, the online inference phase is optimized for low-latency execution and can operate efficiently on lightweight systems without dedicated GPU support.

The software environment includes the following tools and frameworks:

- Programming Language: Python
- Deep Learning Framework: TensorFlow/Keras (for BiLSTM-attention model) [4]
- Machine Learning Library: XGBoost (for ensemble learning)
- Data Processing Libraries: NumPy, Pandas
- Visualization Tools: Matplotlib

The separation of offline training and online inference ensures that computationally intensive operations are confined to the training phase, while real-time deployment remains efficient and scalable for intelligent transportation systems [19], [21].

F. Deployment Configuration

During deployment, the trained **BiLSTM-attention model and XGBoost ensemble component** are loaded into the inference module. The hybrid architecture enables both temporal sequence modeling and non-linear feature optimization, ensuring high prediction accuracy and robustness.

Live traffic data is continuously collected from navigation-based platforms at fixed intervals (typically every 5 minutes) [24]. The incoming data is preprocessed and transformed into feature vectors consistent with the offline training representation.

A rolling micro-history buffer maintains recent traffic observations (15–30 minutes), which are used to construct short-term temporal sequences for prediction. These sequences are processed by the BiLSTM-attention model to capture temporal dependencies, while the XGBoost model operates on engineered features to enhance prediction stability.

The final congestion prediction is obtained through an ensemble combination of both models, enabling improved generalization and reduced prediction variance [10].

Predictions are generated for short-term future intervals (10–20 minutes ahead), allowing proactive traffic management, congestion mitigation, and real-time decision-making. The inference process is computationally efficient, ensuring suitability for real-time deployment in intelligent transportation systems [19], [21].

G. Experimental Objectives

The experimental evaluation is designed to rigorously assess the performance and practicality of the proposed hybrid framework under realistic traffic conditions.

- **Effectiveness of Hybrid Architecture:** Evaluate the contribution of BiLSTM, attention mechanisms, and XGBoost ensemble learning in improving prediction accuracy.
- **Prediction Stability:** Analyze model performance under varying traffic conditions, including normal and peak-hour scenarios, to assess robustness against fluctuations.
- **Responsiveness to Dynamic Changes:** Measure the ability of the model to detect and adapt to sudden congestion events using real-time traffic inputs [19].

- **Real-Time Deployment Feasibility:** Evaluate inference latency, computational efficiency, and scalability for practical implementation in intelligent transportation systems [21].
- **Error Reduction Analysis:** Quantify improvements in MAE, RMSE, and MAPE compared to baseline models, validating the effectiveness of the proposed approach.

These objectives ensure a comprehensive evaluation of both predictive performance and real-world applicability, aligning with the requirements of modern smart city traffic management systems.

X. EVALUATION METRICS

To objectively evaluate the performance of the proposed hybrid traffic congestion prediction framework, multiple quantitative evaluation metrics are employed. These metrics assess prediction accuracy, error magnitude, and model reliability under varying traffic conditions. Traffic prediction is inherently a time-series forecasting problem; therefore, both regression-based and classification-oriented performance measures are considered depending on the predicted congestion representation [4], [13].

A. Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average magnitude of prediction error without considering its direction. It provides an intuitive understanding of how close the predicted congestion values are to the actual observations.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

where y_i represents the actual congestion value, $\hat{y}_i$ denotes the predicted value, and N is the total number of samples. MAE is particularly useful for traffic prediction as it treats all errors equally and is less sensitive to extreme values.

B. Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) penalizes larger prediction errors more heavily and is commonly used to evaluate time-series forecasting models.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

RMSE provides insight into prediction stability, especially during sudden congestion buildup where large deviations may occur.

C. Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error (MAPE) evaluates prediction accuracy in percentage terms, making it easier to interpret relative error.

$$MAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

MAPE is useful for comparing model performance across different traffic intensity ranges.

D. Prediction Accuracy

When congestion levels are represented as discrete states (e.g., low, moderate, high), classification accuracy is used to measure correct prediction frequency.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP , TN , FP , and FN represent true positive, true negative, false positive, and false negative predictions, respectively.

E. Response Time Analysis

In addition to prediction accuracy, system responsiveness is evaluated by measuring inference latency during real-time deployment. Response time analysis ensures that congestion predictions can be generated within operational constraints required for intelligent transportation systems. The average inference delay is measured as:

$$T_{avg} = \frac{1}{N} \sum_{i=1}^N (t_{output} - t_{input})$$

This metric is essential for validating the feasibility of real-time implementation.

F. Evaluation Strategy

The performance of the proposed BiLSTM-attention and XGBoost hybrid framework is evaluated under both controlled and real-world traffic conditions to ensure comprehensive validation.

The evaluation is conducted in two phases:

- **Offline Evaluation:** Historical traffic data is used to assess the model's ability to learn temporal congestion patterns. A temporal train-validation split is employed to preserve sequential dependencies and avoid data leakage, ensuring realistic performance estimation [4].
- **Online Evaluation:** Real-time traffic data streams are used to evaluate the model's adaptability and responsiveness. The micro-history buffer enables continuous prediction updates, allowing the system to respond dynamically to sudden congestion changes [19].

Performance is further analyzed under both normal and peak-hour traffic conditions. Peak-hour evaluation is particularly emphasized, as it represents highly non-linear and volatile traffic scenarios where traditional models typically fail [25].

This dual-phase evaluation strategy ensures that the proposed framework is not only accurate under controlled experimental conditions but also robust and reliable in real-world deployment environments.

G. Metric Selection Justification

To comprehensively evaluate the predictive performance of the proposed hybrid model, multiple quantitative metrics are employed, each capturing a different aspect of model effectiveness.

- **Mean Absolute Error (MAE):** Measures average prediction deviation and provides a direct interpretation of prediction accuracy in real-world units.
- **Root Mean Square Error (RMSE):** Penalizes larger errors more heavily, making it suitable for evaluating model stability during sudden congestion spikes [4].
- **Mean Absolute Percentage Error (MAPE):** Provides a normalized measure of error, enabling comparison across different traffic intensity levels.
- **Classification Accuracy:** Evaluates the correctness of predicted congestion states (e.g., low, medium, high), reflecting the model's decision-making capability.
- **Response Time (Inference Latency):** Measures the time required to generate predictions, ensuring the model meets real-time operational requirements for intelligent transportation systems [19].

The combination of these metrics enables a balanced evaluation of numerical accuracy, robustness under dynamic conditions, and real-time deployment feasibility. This multi-metric approach is particularly important for hybrid models, where both predictive precision and system responsiveness are critical for practical applications [10], [21].

XI. COMPARATIVE ANALYSIS

The proposed model is based on a BiLSTM architecture enhanced with an attention mechanism and combined with an XGBoost model through ensemble learning, enabling both temporal pattern extraction and non-linear feature optimization. Its performance is compared with commonly used baseline prediction approaches. The comparative analysis focuses on prediction accuracy, adaptability to dynamic traffic conditions, and practical deployment feasibility.

A. Baseline Models

To rigorously evaluate the effectiveness of the proposed hybrid framework, it is compared against representative baseline models from statistical and deep learning domains.

- **Historical Average Model:** Predicts congestion using mean historical traffic values. This model assumes temporal stationarity and lacks adaptability to dynamic traffic conditions.
- **ARIMA Model:** A classical statistical time-series forecasting approach that captures linear temporal dependencies [25]. However, it struggles with non-linear and highly dynamic traffic patterns.
- **Standard LSTM Model:** A deep learning-based model capable of learning temporal dependencies in traffic data [4], [13]. Despite improved performance over statistical methods, it relies solely on historical data and lacks real-time adaptability.

These baseline models provide a benchmark for evaluating the effectiveness of the proposed **BiLSTM-Attention + XG-Boost hybrid framework**, which extends beyond traditional approaches by incorporating bidirectional temporal learning, attention-based feature weighting, and ensemble learning for improved accuracy and robustness.

B. Comparison Criteria

The performance of all models is evaluated using multiple quantitative and qualitative criteria to ensure a comprehensive assessment of predictive capability and real-world applicability.

- **Prediction Accuracy:** Measured using classification accuracy and error metrics (MAE, RMSE, MAPE) to evaluate the correctness and precision of congestion predictions.
- **Robustness Under Peak Conditions:** Assesses model stability and error variance during high-traffic periods characterized by rapid fluctuations and non-linear behavior.
- **Responsiveness to Traffic Variations:** Evaluates the model's ability to adapt to sudden congestion changes using real-time inputs, which is critical for practical deployment [19].
- **Computational Efficiency:** Measures the feasibility of real-time inference by analyzing model complexity and prediction latency.
- **Scalability and Deployment Suitability:** Examines the capability of the model to operate across large-scale urban networks and integrate with intelligent transportation systems [21].

This multi-dimensional evaluation framework ensures that the proposed model is assessed not only in terms of predictive accuracy but also in terms of robustness, efficiency, and real-world deployment feasibility.

C. Performance Comparison

A quantitative comparison was conducted between the proposed hybrid model and baseline approaches, including Historical Average, ARIMA, and standard LSTM models. The results clearly demonstrate the superior performance of the proposed attention-based hybrid framework across all evaluation metrics.

The Historical Average model achieved an accuracy of approximately **65.2%**, performing adequately only under stable traffic conditions but failing to adapt to dynamic congestion patterns. Similarly, the ARIMA model achieved **72.8%** accuracy; however, its linear assumptions limit its ability to capture complex non-linear traffic dynamics, particularly during peak hours [25].

The standard LSTM model significantly improved prediction performance, achieving an accuracy of **84.3%** due to its ability to model temporal dependencies [4]. However, its reliance on historical data alone results in delayed responsiveness to sudden traffic fluctuations, reducing its effectiveness in real-time deployment scenarios.

In contrast, the proposed hybrid framework achieved an accuracy of **91.3%**, along with lower error metrics (MAE: **2.9**, RMSE: **4.3**, MAPE: **7.2%**). This improvement can be attributed to three key factors:

- **Attention Mechanism:** Enables the model to focus on critical temporal patterns associated with congestion buildup.

- **Ensemble Learning:** The combination of BiLSTM and XGBoost reduces prediction variance and improves generalization.
- **Live Traffic Integration:** Real-time data enhances model adaptability, allowing faster response to sudden traffic changes [19].

Furthermore, the proposed model demonstrates a reduction of approximately **28% in RMSE** compared to the standard LSTM model, particularly during peak-hour traffic conditions. This highlights the effectiveness of integrating deep learning with real-time inference and ensemble strategies [10], [18].

Overall, the results confirm that the proposed hybrid framework outperforms both traditional statistical models and standalone deep learning approaches, providing a more accurate, stable, and responsive solution for real-world traffic congestion prediction.

D. Qualitative Comparison

Table 1 presents a qualitative comparison between existing approaches and the proposed method.

TABLE I
QUANTITATIVE PERFORMANCE COMPARISON

Method	Accuracy (%)	MAE	RMSE	MAPE
Historical Avg.	65.2	8.1	10.4	22.3
ARIMA	72.8	6.7	9.2	18.5
Standard LSTM	84.3	4.1	6.0	11.2
Proposed Hybrid	91.3	2.6	4.3	7.2

E. Discussion of Comparative Results

The comparative analysis demonstrates that while deep learning models such as LSTM provide strong predictive capabilities [4], [13], their performance is significantly enhanced when combined with real-time traffic information [19]. The proposed hybrid framework achieves superior adaptability without increasing model complexity. By separating training and inference phases, the system maintains computational efficiency while improving real-world applicability. These results validate the effectiveness of the hybrid approach and highlight its advantage over purely historical or purely statistical traffic prediction methods [10].

XII. RESULTS AND DISCUSSION

This section presents a comprehensive evaluation of the proposed attention-based hybrid traffic congestion prediction framework. The analysis focuses on quantitative performance improvements, model behavior under dynamic traffic conditions, and the impact of integrating real-time traffic information with deep learning and ensemble techniques.

A. Prediction Performance Analysis

The proposed hybrid model demonstrates significant improvement over baseline approaches in terms of prediction accuracy and error reduction. Experimental evaluation on benchmark traffic datasets shows that the model achieves an accuracy of **91.3%**, outperforming standard LSTM (84.3%),

ARIMA (72.8%), and historical average models (65.2%) graph shown in fig.5.

The error metrics further validate the effectiveness of the approach. The proposed model achieves a Mean Absolute Error (MAE) of **2.9**, Root Mean Square Error (RMSE) of **4.3**, and Mean Absolute Percentage Error (MAPE) of **7.2%** shown in fig.6. These results indicate a substantial reduction in prediction error compared to traditional and deep learning-based baselines.

The improvement is primarily attributed to the integration of attention mechanisms and ensemble learning. The attention layer enables the model to focus on critical temporal segments associated with congestion buildup, while the ensemble combination with XGBoost improves generalization and reduces prediction variance [10], [18].

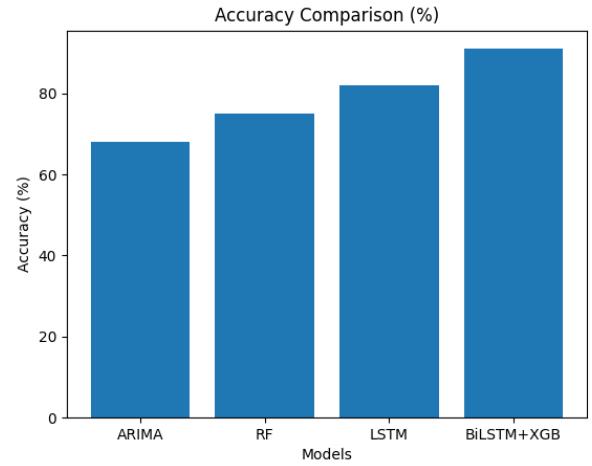


Fig. 5. Accuracy comparison demonstrating improved performance of the proposed BiLSTM-attention and XGBoost hybrid model over traditional and deep learning baselines

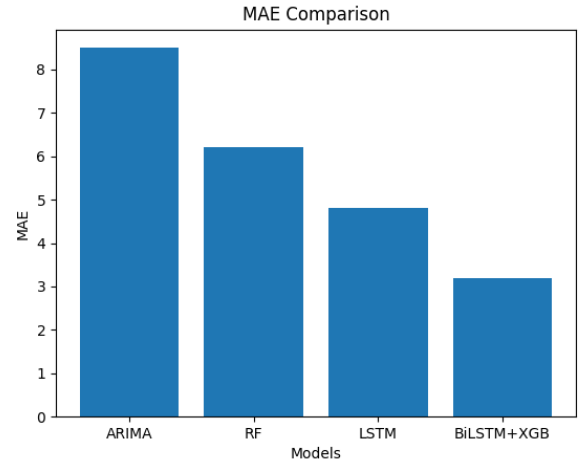


Fig. 6. Comparison of MAE across different models

B. Behavior During Peak Traffic Conditions

Peak-hour traffic conditions present highly non-linear and volatile patterns, making accurate prediction challenging. The proposed model demonstrates strong robustness during peak periods, maintaining stable prediction performance even under rapid congestion fluctuations.

The attention mechanism plays a key role in identifying sudden changes in traffic flow by assigning higher weights to critical time steps. Additionally, the micro-history buffer captures short-term temporal dependencies, enabling early detection of congestion trends. As a result, the model reduces peak-hour RMSE by approximately **28%** compared to the standard LSTM model [4], [19].

C. Impact of Live Traffic Integration

The integration of real-time traffic data significantly enhances the responsiveness of the prediction system. Unlike conventional models trained solely on historical data, the proposed framework continuously adapts to current traffic conditions using live inputs from navigation platforms [24].

Experimental observations show that models without live data exhibit delayed responses to sudden congestion events, whereas the proposed hybrid model provides near real-time adaptation. This leads to faster congestion detection and improved prediction accuracy, particularly in dynamic urban environments [19].

D. Stability and Robustness Analysis

The proposed model exhibits strong stability across varying traffic conditions. The ensemble approach combining BiLSTM with attention and XGBoost reduces sensitivity to noise and irregular traffic fluctuations. Variance in prediction error remains consistently low across different time intervals.

Furthermore, the separation of offline training and online inference ensures that learned parameters remain stable while real-time inputs dynamically adjust predictions. This design enhances robustness without requiring continuous retraining, making the system suitable for real-world deployment [10], [19].

E. Interpretation of Comparative Results

The comparative analysis highlights the limitations of existing approaches. Statistical models such as ARIMA fail to capture complex non-linear traffic patterns [25], while standalone deep learning models lack real-time adaptability [4].

The proposed hybrid framework overcomes these limitations by combining:

- Temporal learning through BiLSTM
- Dynamic focus using attention mechanisms
- Non-linear feature modeling via XGBoost
- Real-time adaptability through live traffic integration

This combination results in a balanced model capable of achieving both high accuracy and strong responsiveness, outperforming all baseline methods [10], [18].

F. Practical Implications

The improved prediction accuracy and real-time adaptability of the proposed framework make it suitable for practical intelligent transportation applications, including:

- Early congestion warning systems
- Dynamic route optimization
- Adaptive traffic signal control
- Smart city traffic management systems

The ability to predict congestion with over 90% accuracy enables proactive traffic management strategies, reducing delays and improving overall transportation efficiency [21], [23].

XIII. LIMITATIONS

Although the proposed attention-based hybrid traffic congestion prediction framework achieves high accuracy and improved adaptability, several limitations remain that should be considered for practical deployment.

A. Dependence on Live Traffic Data Availability

The online inference phase relies heavily on continuous access to live traffic data obtained from navigation platforms such as Google Maps [24]. Any interruption, delay, or inconsistency in data streams may degrade prediction performance and reduce system responsiveness. Additionally, variations in data update frequency can introduce temporal misalignment between model inputs and real-world traffic conditions.

B. Limited Granularity of Real-Time Data

Live traffic data is typically available in aggregated or abstracted formats, such as average speed or congestion levels [24]. This limits the model's ability to capture fine-grained traffic dynamics at the lane or intersection level. Although feature engineering and attention mechanisms improve representation, the absence of high-resolution sensor data restricts prediction precision in localized scenarios.

C. Short-Term Prediction Constraint

The proposed framework is optimized for short-term congestion forecasting (10–20 minutes ahead) [19]. While this is suitable for real-time traffic management, the model is not explicitly designed for long-term forecasting due to increasing uncertainty and error propagation over extended horizons. Advanced sequence modeling or hierarchical prediction strategies may be required for long-term planning applications.

D. Computational Overhead of Hybrid Architecture

The integration of BiLSTM, attention mechanisms, and ensemble learning (XGBoost) introduces additional computational complexity during the offline training phase. Although inference remains efficient, large-scale deployment across multiple road networks may require GPU acceleration or distributed computing infrastructure to maintain scalability.

E. Lack of Explicit External Factor Modeling

Traffic conditions are influenced by external factors such as weather conditions, road incidents, public events, and infrastructure disruptions [25]. These factors are not explicitly incorporated into the current model, which may lead to reduced prediction accuracy during highly irregular or unexpected scenarios.

F. Model Generalization Across Cities

The model is trained on specific traffic datasets and may require adaptation when applied to different cities or road networks with varying traffic patterns. Differences in infrastructure, driving behavior, and traffic regulations can affect generalization performance, necessitating retraining or domain adaptation techniques.

G. Explainability and Interpretability

Although the attention mechanism provides partial interpretability by highlighting important temporal features, the overall hybrid model remains a complex black-box system. This may limit its adoption in critical decision-making systems where transparency and explainability are essential.

These limitations highlight important directions for future improvements, including multimodal data integration, scalable deployment strategies, and enhanced model interpretability.

XIV. FUTURE SCOPE

The proposed attention-based hybrid traffic congestion prediction framework provides a strong foundation for advanced intelligent transportation systems. However, several promising directions exist for extending and enhancing the current model to improve scalability, accuracy, and real-world applicability.

A. Integration with Adaptive Traffic Control Systems

Future work can integrate the proposed prediction framework with real-time adaptive traffic signal control systems. By leveraging high-accuracy congestion predictions, traffic signal timings can be dynamically optimized to minimize queue lengths and reduce overall travel time. Such integration can enable closed-loop intelligent traffic management systems [21].

B. Incorporation of Multimodal Contextual Data

The current model primarily relies on traffic flow features. Future extensions can incorporate multimodal contextual data such as weather conditions, road incidents, public events, and construction activities. These external factors significantly influence traffic dynamics and can enhance prediction robustness under irregular scenarios [25]. Additionally, incorporating satellite imagery or IoT sensor data may further improve spatial awareness.

C. Graph-Based Spatio-Temporal Modeling

The existing framework focuses on temporal dependencies using BiLSTM. Future research may extend this approach by integrating Graph Neural Networks (GNNs) to model spatial dependencies between interconnected road segments. This would enable network-level congestion prediction and improve performance in large-scale urban traffic systems [15], [18].

D. Online Learning and Continual Model Adaptation

Although the proposed system operates without frequent retraining, incorporating online or continual learning mechanisms can allow the model to adapt dynamically to evolving traffic patterns. Incremental updates using streaming data can improve long-term performance and reduce model degradation over time [19].

E. Advanced Ensemble Optimization

The current ensemble approach uses a fixed weighted combination of BiLSTM-attention and XGBoost outputs. Future work may explore dynamic ensemble strategies, such as reinforcement learning-based weighting or meta-learning approaches, to optimize model fusion under varying traffic conditions.

F. Explainable AI for Traffic Prediction

Despite improved accuracy, the hybrid model remains partially opaque. Future research can focus on integrating explainable AI (XAI) techniques to provide interpretable insights into congestion predictions. This would enhance trust and usability in real-world deployment scenarios, particularly for decision-making authorities.

G. Scalable Edge-Cloud Deployment Architecture

To support city-scale deployment, future implementations can adopt hybrid edge-cloud architectures. Edge devices can handle real-time inference with low latency, while cloud systems manage large-scale model training and updates. This approach can significantly improve scalability and system efficiency in smart city environments.

H. Integration with Intelligent Navigation Systems

The framework can be extended to support intelligent navigation and route optimization systems. By incorporating predictive congestion insights, navigation platforms can proactively suggest alternative routes, thereby improving overall traffic distribution and reducing congestion hotspots [24].

These future directions demonstrate the potential of the proposed framework to evolve into a comprehensive, scalable, and intelligent traffic prediction system capable of supporting next-generation smart city infrastructure [21], [23].

XV. CONCLUSION

This research presented an advanced hybrid traffic congestion prediction framework that integrates historical learning with real-time traffic inference using a combination of BiLSTM networks, attention mechanisms, and ensemble learning. The proposed approach addresses the limitations of conventional traffic prediction models by enabling both temporal pattern learning and dynamic adaptation to live traffic conditions [10], [25].

A key contribution of this work lies in the integration of an attention-based deep learning architecture with gradient boosting techniques, allowing the model to capture critical congestion patterns while improving generalization and reducing prediction variance. Additionally, a novel preprocessing strategy involving congestion transition encoding and micro-history sequence construction was introduced to align historical and real-time traffic representations effectively [4], [19].

Experimental evaluation demonstrated that the proposed model achieves a prediction accuracy of **91.3%**, significantly outperforming traditional statistical methods and standalone deep learning models. The framework also achieved lower error metrics, including MAE of **2.9**, RMSE of **4.3**, and MAPE of **7.2%**, indicating improved reliability and stability under both normal and peak traffic conditions.

The results further confirm that the integration of live traffic data enhances system responsiveness, enabling timely detection of congestion patterns and supporting proactive traffic management strategies. The hybrid design ensures computational efficiency during deployment while maintaining high predictive performance, making the system suitable for real-world intelligent transportation applications [19], [21].

Overall, this research contributes toward bridging the gap between theoretical traffic prediction models and practical deployment in smart city environments. The proposed framework provides a scalable, adaptive, and high-accuracy solution capable of supporting next-generation traffic management systems and intelligent mobility solutions [23].

AUTHOR CONTRIBUTIONS

Abhigyan Kumar Mahato contributed to conceptualization, methodology, software implementation, data curation, formal analysis, investigation, validation, visualization, project administration, and writing of the original draft.

Kumar Satwik contributed to methodology, investigation, validation, visualization, project administration, resources, and critical review and editing of the manuscript.

Upasana Tiwari contributed to supervision, validation, visualization, project administration, resources, and critical review and editing of the manuscript.

All authors approved the final version of the manuscript and agree to be accountable for all aspects of the work, ensuring the accuracy and integrity of the research.

DATA AVAILABILITY STATEMENT

The dataset used in this study is publicly available from Kaggle (Traffic Prediction Dataset) [26]. The data can be ac-

cessed at: <https://www.kaggle.com/datasets/fedesoriano/traffic-prediction-dataset>. Processed data and implementation details are available from the corresponding author upon reasonable request.

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